

JEE MAIN REVISION MASTER: DUAL NATURE OF RADIATION 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 25 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Dual Nature Of Radiation - Core Application	25 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Dual Nature Of Radiation - Core Application (Very High Yield — 25 PYQs)

Core Principle: Key problem pattern in Kcet Dual Nature Of Radiation.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Variation of photoelectric current with anode potential is shown below. Choose the correct option (V_0 = stopping potential)..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Work function of the metal is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "If we consider an electron and a photon of same de-Broglie wavelength, then they will have same..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Variation of photoelectric current with anode potential is shown below. Choose the correct option (V_0 = stopping potential)

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: In the photoelectric effect, photoelectric current changes with anode potential in this way: 1. At zero anode potential, the current is still present. This happens because the emitted photoelectrons already have some kinetic energy, so some of them can reach the anode even without extra attraction. 2. When the anode is given a negative potential, it repels the photoelectrons. So, fewer electrons reach the anode and the current decreases. 3. At a particular negative potential, even the fastest photoelectrons cannot reach the anode. The current becomes zero at this point. This potential is called